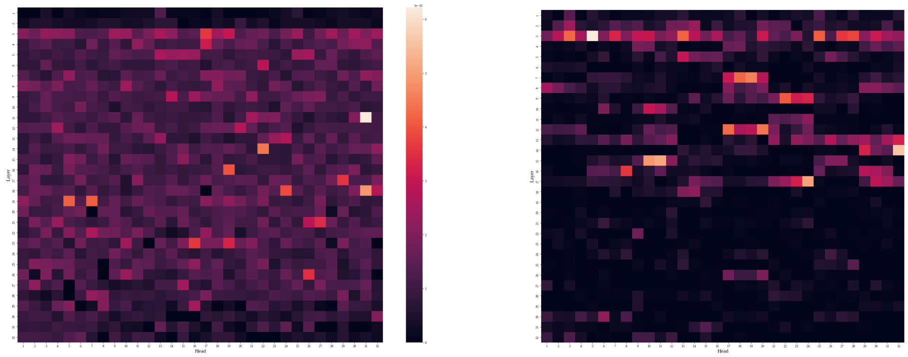
Table 9: MedQA Question Answering GPT4-Judge

| Cache Budget / Fastgen Attn Recovery Frac (%) | Strategy | Similarity to GT | Coherence | Faithfulness | Helpfulness |
|---|---|---|---|---|---|
| 10 | HASHEVICT | 1.970 | 3.517 | 2.665 | 2.547 |
| | $L_2$ | 1.103 | 1.695 | 1.639 | 1.283 |
| | $H_2O$ | 2.138 | 3.206 | 2.594 | 2.416 |
| | Scissorhands | 2.144 | 3.202 | 2.580 | 2.402 |
| 30 | HASHEVICT | 2.511 | 4.415 | 3.533 | 3.613 |
| | $L_2$ | 1.939 | 3.633 | 2.942 | 2.843 |
| | $H_2O$ | 3.428 | 3.818 | 3.608 | 3.276 |
| | Scissorhands | 3.406 | 3.850 | 3.602 | 3.286 |
| 50 | HASHEVICT | 3.022 | 4.730 | 4.139 | 4.254 |
| | $L_2$ | 2.850 | 4.511 | 3.797 | 3.950 |
| | $H_2O$ | 2.938 | 3.632 | 3.280 | 2.762 |
| | Scissorhands | 2.918 | 3.634 | 3.308 | 2.748 |
| 70 | HASHEVICT | 3.232 | 4.809 | 4.292 | 4.434 |
| | $L_2$ | 3.194 | 4.755 | 4.235 | 4.385 |
| | $H_2O$ | 2.414 | 3.396 | 2.958 | 2.178 |
| | Scissorhands | 2.554 | 3.454 | 3.098 | 2.328 |
| 90 | HASHEVICT | 3.291 | 4.839 | 4.355 | 4.507 |
| | $L_2$ | 3.265 | 4.818 | 4.318 | 4.458 |
| | $H_2O$ | 2.400 | 3.232 | 2.830 | 2.016 |
| | Scissorhands | 2.404 | 3.346 | 2.980 | 2.098 |
| 50 | Fastgen | 1.002 | 1.004 | 1.006 | 1.000 |
| 60 | | 1.005 | 1.004 | 1.005 | 1.000 |
| 70 | | 1.008 | 1.014 | 1.014 | 1.008 |
| 80 | | 1.620 | 2.783 | 2.270 | 1.512 |
| 90 | | 2.356 | 3.242 | 2.748 | 1.870 |
| 100 | Full | 3.337 | 4.817 | 4.342 | 4.500 |

Table 10: **LSH Hash Dimension Ablation.** We assesses GSM8K Question Answering performance for different LSH dimensions. The cache budget is fixed at 50%. LSH dimension does not significantly impact performance. Small LSH dimensions slightly outperform larger LSH dimensions.

| LSH Dim | BERTScore F1 | Rouge L | GPT4 Judge | Compression Ratio | Cache Memory (GB) |
|---|---|---|---|---|---|
| 4 | 0.8807 | 0.3974 | 4.3833 | 0.3728 | 2.8062 |
| 8 | 0.8802 | **0.3975** | **4.4113** | 0.3734 | 2.8355 |
| 16 | **0.8807** | 0.3972 | 4.3753 | 0.3716 | 2.8941 |
| 24 | 0.8802 | 0.3951 | 4.3733 | 0.3711 | 2.9527 |
| 32 | 0.8796 | 0.3926 | 4.3220 | 0.3710 | 3.0113 |
| 64 | 0.8797 | 0.3900 | 4.2333 | 0.3702 | 3.2456 |



(a) ALR using LSH ranking (b) ALR using $L_2$ ranking

Figure 8: **Attention Loss Ratio (ALR).** We compare how the eviction strategy of HASHEVICT and the $L_2$ method [Devoto et al., 2024] affects the ALR per equation 7. Our tested model is Llama3-8B-Instruct, which contains 32 heads and 32 attention layers. Cell $(i, j)$ depicts the ALR of head $i$ in attention layer $j$. A darker score indicates a lower ALR. The $L_2$ method exhibits extremely low ALR, thus indicating exclusive preference for high-attention tokens. HASHEVICT prefers to select medium-to-high attention tokens.

Table 11: Average Pearson correlation between attention scores and inverted average Hamming distances per projection length, computed for 50 randomly selected prompts from GSM8k. Higher projection lengths have stronger correlations.

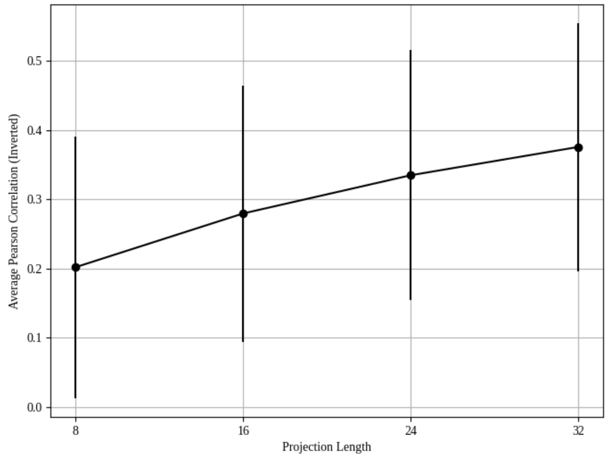
| Projection Length | Mean | Standard Deviation |
|---|---|---|
| 8 | 0.2017 | 0.1890 |
| 16 | 0.2793 | 0.1852 |
| 24 | 0.3345 | 0.1806 |
| 32 | 0.3754 | 0.1792 |

- **LSH and Token Similarity:** LSH tended to group tokens together that are similar across dimensions, producing lower Hamming distances. Tokens with very high attention scores may only have strong associations for a relatively small subset of dimensions, which may not always be captured effectively by LSH.

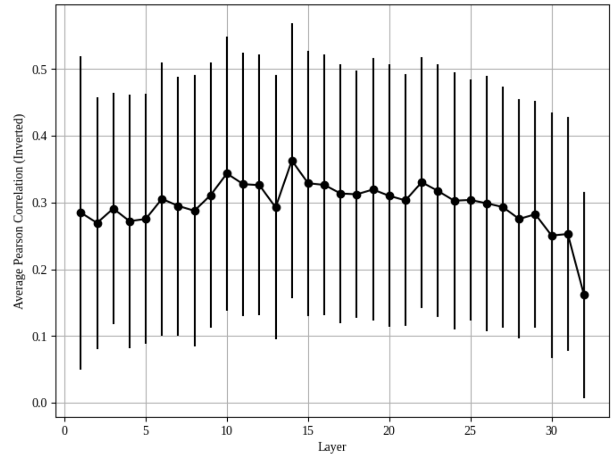
### E.3 ALR Computation Methodology

We compute the Attention Loss Ratio (ALR) for each layer $l$ and head $h$ as follows:

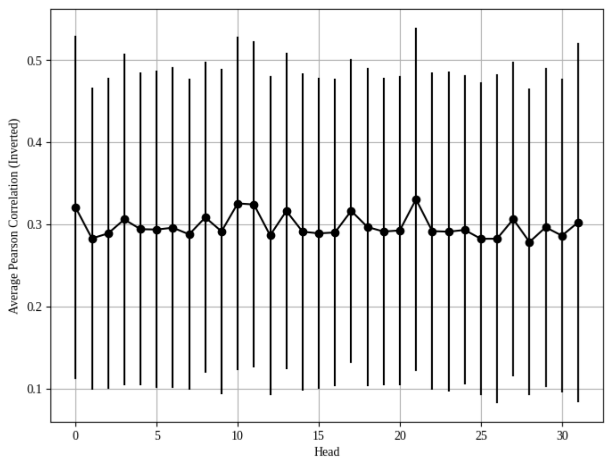
1. **Data Capture** During the model's forward pass, we capture the necessary data for analysis:
   - **Attention Probabilities** $a_{l,h} \in \mathbb{R}^{n \times n}$: The attention scores between queries and keys.
   - **Key Norms** $\|\mathbf{k}_{l,h,p}\|_2$: The $L_2$ norms of key vectors at each position $p$.
   - **Key and Query Vectors** $\mathbf{k}_{l,h,p} \in \mathbb{R}^d$ and $\mathbf{q}_{l,h,p} \in \mathbb{R}^d$: Used for LSH ranking.
2. **Mean Attention Scores** For each token position $p$, we compute the mean attention score across all positions it attends to:
$$\bar{a}_{l,h,p} = \frac{1}{n} \sum_{q=1}^{n} a_{l,h,p,q}. \tag{8}$$
3. **Ranking Methods**
   - **Ideal Attention-Based Ranking** Rank positions in ascending order of $\bar{a}_{l,h,p}$ (from lowest to highest attention score).
   - $L_2$ **Norm Ranking** Rank positions in descending order of the key norms $\|\mathbf{k}_{l,h,p}\|_2$.
   - **LSH Ranking** Apply Locality-Sensitive Hashing (LSH) to key and query vectors using random projections, compute Hamming distances, and rank positions in ascending order of the average Hamming distance.
4. **ALR Calculation** For each $m$ from 1 to $n$, compute the cumulative attention losses: This allows us to quantitatively compare how well different ranking methods (e.g., $L_2$ norm and LSH ranking) approximate the



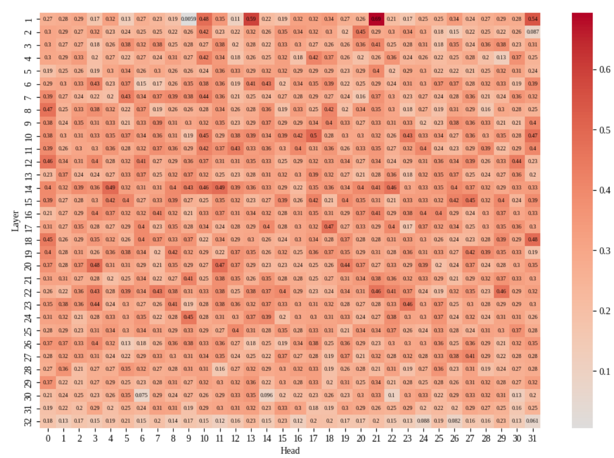

(a) **Correlations for varying LSH dimension.** We study the Pearson correlations between attention scores and the inverted average Hamming distances, computed over 50 randomly selected prompts from GSM8K, as a function of projection length for Llama-3-8B-Instruct. The tested projection lengths are 8,16, 24, and 32. The error bars indicate the standard deviation. Correlation strengthens as projection length increases.




(b) **Correlations by layer.** We measure the Pearson correlations between attention scores and the inverted average Hamming distances for each transformer layer in Llama-3-8B-Instruct computed over 50 randomly selected prompts from GSM8K. Error bars indicate standard deviation. The final three layers have the weakest correlations.




(c) **Correlations by head.** We study the Pearson correlation between attention scores and the inverted average Hamming distances for each head in Llama-3-8B-Instruct computed over 50 randomly selected prompts from GSM8K. Error bars indicate standard deviation. There is minimal variation between heads.




(d) **Correlation Heat Map.** We examine the average Pearson correlation between attention score and the inverted average Hamming distances (LSH ranking) across all layers and attention heads of Llama-3-8B-Instruct. As attention mass tends to concentrate over a few tokens [Gupta et al., 2021, Sheng et al., 2023], the slightly-weak, but positive correlation indicates the LSH ranking is selecting medium-to-high-attention tokens.

Figure 9: **Correlations of Attention and Inverted Hamming Distances**

ideal scenario where the least important KV pairs (those with the lowest attention scores) are dropped during cache compression.

$$L^m_{l,h} = \sum_{i=1}^{m} \bar{a}_{l,h,\pi(i)}, \tag{9}$$

$$L^m_{l,h,\text{ref}} = \sum_{i=1}^{m} \bar{a}_{l,h,\sigma(i)}, \tag{10}$$